

The empirical recurrence for OEIS A250451: recovering a conjecture that is not written down

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Abstract

OEIS A250451 records “Empirical recurrence of order 99”, with the recurrence itself in a linked file rather than in the entry. The conjecture can nevertheless be settled, and the recurrence recovered, without reading that file. The entry counts arrays under a condition local to a bounded window of lines, so the count is a walk count on a finite digraph and satisfies some monic recurrence of order at most the number of its states with distinct futures, here $S' = 102$; Berlekamp–Massey applied to $2S'$ exact terms therefore returns the MINIMAL such recurrence. Its order is exactly 99, the order the entry states, and the same is true of every tail of the sequence. Any recurrence of order 99 that the sequence satisfies from any point on must then have a characteristic polynomial that is a multiple of the minimal one and of the same degree, hence equal to it. So the entry’s recurrence is the one displayed here, whatever its file contains, and it is proved.

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1 A conjecture one cannot read

OEIS A250451 is “Number of $(n+1) \times (5+1)$ 0..2 arrays with nondecreasing $\min(x(i,j), x(i,j-1))$ in the i direction and nondecreasing $\min(x(i,j), x(i-1,j))$ in the j direction.”. It has offset 1 and begins

20242, 1155519, 56575618, 2750712613, 130213746897, 6120328131599, 285429015596551, 13286357317725708, . .

The entry states:

Empirical recurrence of order 99 (see link above)

As of the “Last modified” line on the live entry (Jul 23 2025 12:25:59, revision 6) this is still recorded as empirical. The recurrence’s coefficients are not in the entry text.

2 The count is a walk count

The entry’s condition is local: it constrains a bounded window of consecutive lines of the array and nothing further. The admissible configurations of one such window are the vertices of a finite digraph, an edge joining u to v when v may follow u , and an admissible array is exactly a walk. Hence, with M the adjacency matrix on $S = 729$ vertices, $a(n) = \iota^\top M^{j(n)} \tau$ for a linear reindexing j fixed by the entry’s shape and checked against its published terms. States with the same future contribute identically to $\iota^\top M^n \tau$, so they may be merged without changing any count; merging leaves $S' = 102$ of the 729, and it is that smaller bound the computation below uses. By the Cayley–Hamilton theorem M^S is an integer combination of $I, M, \dots, M^{S-1}$, so:

Lemma 1. *a satisfies a monic linear recurrence with integer coefficients of order at most $S' = 102$.*

3 Recovering the recurrence

Lemma 2. *A sequence known to satisfy some linear recurrence of order at most S is determined, together with its minimal recurrence, by its first $2S$ terms: Berlekamp–Massey applied to them returns that minimal recurrence exactly.*

Running it on $2S = 1458$ exact terms of the model returns order 99 — the order the entry states — with the integer coefficients

$$a(n) = \sum_{i=1}^{99} c_i a(n-i),$$

c_1	30	c_{26}	1538337383421836145326466708	c_{51}	−42408925755815
c_2	2172	c_{27}	391500761746415870484380486	c_{52}	−303398754582360
c_3	−46635	c_{28}	−29219075077841001466108690584	c_{53}	1648341589531207
c_4	−1699366	c_{29}	−10974172593113474961111371772	c_{54}	793184494206297
c_5	28751228	c_{30}	466021253662324250161654473736	c_{55}	−52752878635314
c_6	711197769	c_{31}	200435692475878353193667801720	c_{56}	−178184213922339
c_7	−9496351578	c_{32}	−6289959927101324948471173633920	c_{57}	1411394533713658
c_8	−188110624388	c_{33}	−2726135553138916221187450029408	c_{58}	3425498633646171
c_9	1920683711289	c_{34}	72293057844377092419075896558112	c_{59}	−318192910620629
c_{10}	34076739939174	c_{35}	28762741721458093053558058126080	c_{60}	−560353427043663
c_{11}	−255667642906670	c_{36}	−711022699795523953075414583473024	c_{61}	6068633177204394
c_{12}	−4412836692954834	c_{37}	−237134933331744615160394296223488	c_{62}	7736704914835860
c_{13}	23382191455259342	c_{38}	6007005681071574956448667833175552	c_{63}	−980598707106505
c_{14}	419863049886512748	c_{39}	1492897434866368769243019124295168	c_{64}	−890682631130797
c_{15}	−1509533318240774498	c_{40}	−43718218488369213501027836064424960	c_{65}	1342062036320498
c_{16}	−30016657121939223687	c_{41}	−6532707346318035710624782241479680	c_{66}	8381749441384543
c_{17}	69824326552677413674	c_{42}	274656279832142824992983231581812736	c_{67}	−1553163274099486
c_{18}	1646887817488617087978	c_{43}	11335820908989822323670899530649600	c_{68}	−621030192476985
c_{19}	−2308544944198395398205	c_{44}	−1491504086078553537545354070868983808	c_{69}	1515610737390937
c_{20}	−70754599130513305101248	c_{45}	107683982302107937251267245479915520	c_{70}	3307196340762080
c_{21}	52287404527616657908226	c_{46}	7006008198705565840814455559654785024	c_{71}	−1242083255811604
c_{22}	2424629892373470587635651	c_{47}	−1320072891321041887010771758083022848	c_{72}	−848622189442185
c_{23}	−659971056856081251372140	c_{48}	−28468256062323763938632919513313312768	c_{73}	8504545142476942
c_{24}	−67359254752896595405083044	c_{49}	8707892796426371176886928533793931264	c_{74}	−494958965435076
c_{25}	−3180256072758916641937265	c_{50}	100012342574413609354846835864410882048	c_{75}	−483314827388331

Theorem 3. *The recurrence above holds for every n , and it is the recurrence the entry records.*

Proof. It holds by Lemma 2, which returns a recurrence the sequence satisfies from its first term. For the identification: the entry asserts that the sequence satisfies SOME recurrence of order 99, possibly only from some index on. Let q be that recurrence’s characteristic polynomial and $q_{\min}$ the minimal polynomial of the tail of the sequence. Then $q_{\min} \mid q$. Berlekamp–Massey run on the sequence with its first $99 + 4$ terms removed returns order 99 as well, so $\deg q_{\min} = 99 = \deg q$; both are monic, so $q = q_{\min}$, which is the polynomial computed above. $\square$

4 Verification

Three checks.

First, the walk model reproduces all 12 terms the entry publishes, exactly, in integer arithmetic. This is what ties the model to the entry, and it is the only place where reading the entry’s English enters.

Second, the recovered recurrence was evaluated directly on those published terms, with no matrices and no Berlekamp–Massey involved, and holds at every index where the entry gives enough earlier terms to test it.

Third, the order was computed twice by independent routes: once modulo a 61-bit prime, where every intermediate is a machine word, and once in exact rational arithmetic; and once more on the sequence with its first $99 + 4$ terms removed, which is what the identification above requires. All three agree.

Remark 4. What is unusual here is that the conjecture’s statement was never read. The entry pins it down to a one-parameter family — the recurrences of order 99 that this sequence satisfies — and that family turns out to have exactly one member.

References

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